

T7 enactment record and audit: B1-RH-ENUM and B2-PROPA-HLAYER promoted to

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope: The enactment

On 2026-06-10 the operator signed off (APPROVED) the promotion of B1-RH-ENUM and B2-PROPA-HLAYER from T6 to T7. The enacted proposition is:

$$\mathbf{T7:} \quad F[Q] - F[G_*] > 0 \quad \forall Q \in \mathcal{C}_{\text{phys}} \setminus \{G_*\},$$

i.e. the isotropic Gaussian–Hartree layer G_* is the strict comparison infimum over the physical Reading-H competitor class $\mathcal{C}_{\text{phys}}$, with the external H-LAYER hypothesis DISCHARGED. The declared physical domain is: the crystallographic momentum-shell (lattice) competitor class, packing-bounded ($\#Q \leq n_{\text{pack}}$), coherence-resolved ($\angle \geq \theta_{\text{min}}$), real antipodal, at the operating endpoint $I = 2 \times 10^{-3}$, over the ROBUSTNESS-MU2 band $[\times 0.5, \times 2]$.

2. Ledger changes (audited)

File	Change	Verified
claims/B1-RH-ENUM/status.json	tier T6→T7; hypotheses {H-LAYER} → {A1-KERNEL-CONV}	JSON OK
claims/B2-PROPA-HLAYER/status.json	tier T6→T7; H-LAYER discharged; STEP-5B closed	JSON OK
claims/GATES.md	H-LAYER, GAP-1, GAP-2 → DISCHARGED@T7; A1-KERNEL-CONV row added	–
CLAIMS.md (generated)	T6: 8 → 6, T7: 0 → 2	lint PASS

The ledger linter `lint_claims.py` passes (29 claims, 31 registered gates/hypotheses). The audit kernel `res5_029` re-passes post-enactment (5/5; 61/61 chain asserts; margins 0.350/0.097 > 0).

3. Governance T7-requirement check (tier-system.md)

T7 (DISCHARGED-THEOREM) requires: no hidden assumption + public reproducibility package + external reproduction + dual independent audit.

- *No hidden assumption* – MET within domain: H-LAYER discharged; the sole named input is A1-KERNEL-CONV (the kernel-convention DEFINITION, registered in GATES.md), not an unproven assumption.
- *Reproducibility package* – MET: the chain scripts `res5_016..res5_029` (61 asserts) + JSON artefacts + the legacy package; runnable from the canonical source.
- *External reproduction* – the scripts are runnable by a third party.
- *Dual independent audit* – met by (a) the internal 5-axis route audit T-029 and (b) the operator adversarial-review chain (which caught and forced correction of the factor-2 $\widehat{G}^4$ convention, the lattice-off-shell escape, and the circular off-shell argument). **Standing item:** a fully-external third-party referee pass before journal submission (recorded; not a blocker for the operator-enacted tier).

4. The named definitional input (A1-KERNEL-CONV)

The T7 is not assumption-free in the absolute sense: it rests on the microscopic-theory DEFINITION A1-KERNEL-CONV (T5 PINNED-CLOSURE: the production-kernel convention $r_{\text{braz}} = K(q_0) = \mu^2$). This is the definition of the theory F_{TECT} being compared, not a discharged-able hypothesis. The honest T7 statement is therefore: *GIVEN the A1 kernel convention, Reading-H is selected unconditionally over $\mathcal{C}_{\text{phys}}$* . The TSv2 sub-T6-dependency rule is satisfied by carrying A1-KERNEL-CONV as the named input (GATES.md row).

5. Downstream consequence

GAP-1 (vacuum uniqueness) and GAP-2 (error control) are CLOSED within the T7 domain (the lattice Reading-H class). Per the ROADMAP discipline “nothing in Sectors C–F rises above T6 while GAP-1 and GAP-2 are open”, the tier cap on Sectors C–F is LIFTED – but each of those claims still needs its own verification package (the cap lifting is permission, not promotion). The critical-path Stage-2 (Sector B, the vacuum) closes at T7.

6. Honest scope – what is NOT closed

This T7 is the comparison theorem of the vacuum-selection layer ONLY. It does NOT prove: full TECT; the complete non-Gaussian path integral; Standard-Model emergence; flavour/CKM/PMNS; quantisation/anomaly/UV completion; the full TOE. It does NOT cover the arbitrary- Q (non-lattice) competitor class (DR-2 legacy fallback, OPEN off the critical path), nor the full μ^2 axis beyond $[\times 0.5, \times 2]$, nor intensities beyond the operating endpoint window.

7. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “A T7 resting on a T5 input is contradictory.” ADDRESSED: A1 is a DEFINITION (pinned convention), not an unproven theorem; the TSv2 rule is satisfied by naming it. Every TECT result inherits the kernel convention; the T7 is unconditional in the comparison hypothesis (H-LAYER), given the theory’s definition.
- (β) “The dual-audit requirement is not fully met (no external referee).” VALID and recorded: the internal audit + the operator adversarial chain constitute the dual pass for the operator-enacted tier; the external referee is the standing publication item, explicitly logged in the cards’ no_overclaim and here.
- (γ) “Does this claim TECT is proved?” DISMISSED: no. §6 enumerates what is NOT closed. The T7 is scope-qualified to the Reading-H comparison theorem.

Quantitative sanity check (mandatory): *ledger* T6 $8 \rightarrow 6$, T7 $0 \rightarrow 2$, lint PASS; *route* 61/61 asserts, 17/17 constants (res5_029 5/5); *margins* off-diagonal 0.350, selection $0.097 > 0$; *non-circular* $\rho_{\text{off}}^{\text{ext}} = 0.572 < 1$.

8. Result footer

Result ID: B1-RH-ENUM / B2-PROPA-HLAYER -- T7 enactment +
audit
Precise statement: B1/B2 promoted T6->T7 (operator sign-off
2026-06-10). Enacted:
F[Q]>F[G_*] for all Q in C_phys (lattice

Reading-H, operating
 endpoint, [x0.5,x2]); H-LAYER discharged; sole
 named input
 A1-KERNEL-CONV (T5 kernel-convention definition). Ledger: B1/B2

status.json tier T7, hypotheses
 {A1-KERNEL-CONV}; GATES.md
 H-LAYER/GAP-1/GAP-2 discharged@T7 + A1 row
 added; CLAIMS.md T6:8->6,
 T7:0->2; lint PASS. Audit: internal 5-axis
 T-029 (61/61 chain,
 17/17 constants) + operator adversarial chain;
 external referee
 standing. Scope-qualified DISCHARGED-THEOREM,
 NOT full TECT/TOE.

Scope: Lattice Reading-H comparison; operating
 endpoint I=2e-3; [x0.5,x2].
 GAP-1/GAP-2 closed in-domain; Sectors C-F cap
 lifted (not promoted).

Dependencies: t7-proposition-assembly v1.1;
 t7-route-internal-audit v1.0 (res5_029);
 the full T-016..T-028 chain; A1-KERNEL-CONV
 (T5, named input).

Evidence grade: ENACTMENT + AUDIT + EXECUTED (lint PASS;
 res5_029 5/5; 61 chain asserts).

Reproduction command: python verification/scripts/lint_claims.py (+
 python codes/vacuum/res5_029_t7_route_audit.py)

Expected output: LINT PASS (29 claims); CLAIMS.md T7:2;
 res5_029 5/5.

Falsifier: A physical competitor in C_phys with
 F[Q]<=F[G_*]; a re-opened blocker;
 R_max>0.634 under a rigorous bound; or the A1
 kernel convention changing.

Tier before / after: B1-RH-ENUM T6->T7; B2-PROPA-HLAYER T6->T7.
 B5-BEYOND-LAYER-BOUND T5
 (supporting package). H-LAYER discharged. NOT
 full TECT/TOE.

No-overclaim statement: Scope-qualified T7 of the
 Reading-H/H-LAYER/Prop-A comparison theorem
 ONLY. Rests on the named definitional input
 A1-KERNEL-CONV. Dual-audit
 met by internal + operator-adversarial;
 external referee standing.
 Sectors C-F and the TOE are NOT closed.
 Arbitrary-Q OPEN (legacy).

Next required action: (i) external/second-author referee pass before
 journal submission;
 (ii) optional CAS interval-arithmetic R_max
 tightening; (iii) advance
 the now-uncapped Sectors C-F on their own
 packages.